Nicholas Channg

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Education

Cornell University — Ithaca, NY
B.A. Computer Science, Minors in Artificial Intelligence, Data Science | GPA: 3.72/4.0 — *Expected Graduation, May 2027*

- **Awards:** 3x Hackathon Winner, Y Combinator AI Startup School 2025 (1 of 2,500 selected)

Experience

Google — Sunnyvale, CA
Incoming Software Engineer Intern, Google Cloud Dataplex — *Summer 2026*

Amazon — New York, NY
Software Engineer Intern, Amazon Ads — *Sep 2025 – Dec 2025*

- Built and owned asynchronous Java and Python backend services in a distributed AWS video ads processing pipeline handling 80K+ jobs weekly, designing fault-tolerant workflows across SQS, ECS, Lambda, DynamoDB, and S3
- Developed a production AI microservice that performs multimodal video analysis via AWS Bedrock VLM/LLM, generating optimized ad titles/descriptions and improving text-asset relevance for 98% of video ads
- Integrated a video assembly service into the pipeline by building a Lambda-triggered ECS execution path and S3/DynamoDB asset storage, enabling secure cross-account and cross-service access of final video assets

Tesla — Fremont, CA
Software Engineer Intern, Charging Team — *May 2025 – Sep 2025*

- Developed scalable REST API endpoints in the Tesla Mobile App backend with Node.js, Express.js, and TypeScript, orchestrating vehicle/charging microservice calls to power real-time user-visible data for 3.4M daily users
- Implemented a Redis caching layer to store Spark API responses for supercharger congestion/pricing data, reducing redundant upstream microservice calls and improving Tesla Mobile App response latency by 10%
- Developed an agentic AI QA assistant with Python that automated mobile app test-account setup (e.g. region switching, vehicle linking) via orchestrated microservice API calls, improving QA throughput by 25%

Flow — Remote
Software Engineer Intern — *Jan 2024 – May 2024*

- Developed REST APIs using Flask and Python to retrieve and process data from Flow's AI models, automating customer prospecting and increasing sales representative call volume by 300% (30 → 120 calls/day)
- Reduced application latency by 15% by optimizing PostgreSQL queries; replaced nested SELECTs with JOINs, added B-tree indexes on high-cardinality columns, and batched execution to minimize database calls
- Integrated backend services into CI/CD pipelines using GitHub Actions, ensuring automated testing and deployment

Projects

ElectAI | *Python, Scikit-learn, Flask, React, TypeScript, TailwindCSS, NumPy, Pandas* — Jan 2025

- Engineered a full-stack ML web app to predict voter turnout using historical behavioral data patterns to model participation likelihood using a Flask REST API to retrieve data from the model and a React frontend
- Built a data preprocessing pipeline with Python, NumPy, and Pandas to clean 20 years of election data, enabling accurate regression modeling with Scikit-learn and achieving high accuracy (RMSE: 5.35, MAE: 4.22)

MathGPT | *React, TypeScript, HTML, CSS, Microsoft Phi-3 API* — July 2024

- Built an AI Math Tutor Chrome extension using Microsoft's Phi-3 LLM, React, and TypeScript that provides personalized math tutoring to 100+ students, featuring structured prompts and step-by-step AI solutions
- Reduced latency by 67% (30s → 10s) by dynamically adjusting user input via prompt engineering and implementing conditional API calls, eliminating redundant requests to the LLM and timeout issues

Technical Skills

Languages: Java, Python, C, C++, TypeScript, JavaScript, SQL
Frameworks: Flask, FastAPI, Node.js, Express.js, React, Next.js
Cloud/Infrastructure: AWS, Redis, PostgreSQL, Docker, CI/CD
Libraries/Tools: Scikit-learn, NumPy, Pandas, OpenAI API, TailwindCSS, Git, Postman